Online Appendix to "Computing the Exact Distribution of Predictive Regression"

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This appendix collects supplementary derivations and implementation details. The first four sections record what Section 5 of the paper leaves to the code: the substitutions by which the moment generating function is evaluated through the assembly of Sections 3 and 4, the settings and cross-checks of the derivative evaluators, the quadrature maps and node counts, and the measured cost and validation. The derivations themselves are in the paper and are not repeated. Section OA.5 derives the behavior of the density of R^2 at the origin, stated in Section 2.3 of the paper. Equation and section numbers of the form (x.y) refer to the paper.

OA.1 Evaluating the mgf in the released code

Section 5.1 of the paper establishes that the moment generating function

$$\Psi(\gamma, s, t) = |V_y|^{-\frac{1}{2}} \, |W(\gamma, s, t)|^{-\frac{1}{2}}, \qquad W(\gamma, s, t) \;=\; V_y^{-1} - 2\gamma A + 2sB + 2tC, \tag{OA.1}$$

is the determinant of Sections 3 and 4 of the paper at real rather than imaginary arguments, so that the entire $O(1)$ reduction applies verbatim and the branch analysis is not needed at all. This section records how that substitution is carried out in the released code and what the implementation is checked against.

The evaluator `qfmgf_logpsi_f.m` obtains $\log \Psi$ by calling the assembly of the characteristic function. Writing (s_c, t_c) for the arguments at which that assembly expects the transforms of $y'Ay$ and $y'Cy$, and c_0 for the coefficient with which $y'By$ enters the diagonal, the substitution is

$$s_c = -\mathrm{i}\,\gamma, \qquad t_c = \mathrm{i}\,t, \qquad c_0 = \frac{2s}{T}, \tag{OA.2}$$

after which the Schur elimination of the return block, the primitives of the tridiagonal Toeplitz core, the Sherman–Morrison step for the demeaning term and the 2×2 border factor are those of the paper, unchanged. In the $y = (r_1, \ldots, r_T, z_0, \ldots, z_{T-1})'$ coordinates the matrices A, B and C carry no dependence on the slope, which is why the return block of W stays diagonal and the elimination is available at all. The companion routine `qfmgf_setup.m` evaluates the three scale constants $\mathbb{E}[y'Ay]$, $\mathbb{E}[y'By]$ and $\mathbb{E}[y'Cy]$ by summing their geometric series in closed form, so that nothing of size T is built or stored and the setup is $O(1)$ as well. All computations are carried out in logarithms, the modulus and the phase being

accumulated separately, so that the frequent and harmless underflow of Ψ deep in the tilted region never produces a NaN.

A second evaluator, `qfmgf_logpsi.m`, factors W directly: in the interleaved ordering $(u_1, z_0, u_2, z_1, \ldots)$ it is banded with bandwidth three up to the exact rank-two demeaning correction, and a banded LDL' factorization vectorized across quadrature nodes gives $\log \Psi$ in $O(T)$ operations per node. This route is retained, selected by `engine = 'banded'`, purely as an independent check on the closed-form assembly, and `test_o1.m` exercises the comparison. Against a dense determinant at $T \leq 9$ both agree to 4×10^{-15}; against each other they agree to a relative 2×10^{-12} at $T = 120$ and about 10^{-9} at $T = 960$ and beyond. The residual at large T is a property of the banded reference, not of the closed-form assembly: the factorization accumulates rounding over $O(T)$ pivots and recovers the near-confluent argument by subtraction, whereas the closed-form route, which forms $p + 2q$ in closed form, is accurate to 5×10^{-12} or better against arbitrary-precision spectral references at every sample size tried (Section 3.3 of the paper).

All computations are carried out in logarithms, the modulus and the phase being accumulated separately, so that the frequent and harmless underflow of Ψ deep in the tilted region never produces a NaN.

OA.2 Derivatives of the mgf: settings and cross-checks

Section 5.2 of the paper derives the two devices the code uses—a complex step for the first derivative, Taylor-mode (jet) arithmetic above it—together with the precautions the tridiagonal Toeplitz primitives require and the accuracy attained. What follows are the implementation settings and the routine that checks them.

Complex step. The released code takes the step

$$h_s = 10^{-60} \cdot \gamma_{\text{scale}}, \qquad \gamma_{\text{scale}} = \big(\mathbb{E}[y'By]\,\mathbb{E}[y'Cy]\big)^{-1/2},$$

γ_{scale} being the natural scale of the tilt, with both expectations supplied by `qfmgf_setup.m`. A single evaluation of $\log \Psi$ at that complex argument delivers both $\log \Psi(\gamma_0)$, as its real part, and the first derivative, as its imaginary part divided by h_s.

Taylor mode. The series assembly is `qfmgf_dj.m`, reached through the driver `qfmgf_dp.m`. Its variant `qfmgf_djs.m`, which carries the series in the second argument, supplies the s-derivatives that the odd-moment representation of the t-ratio requires.

Cauchy-circle cross-check. An independent evaluator based on Cauchy coefficients on a small circle, taken by the FFT (Lyness and Moler, 1967), is retained in the released code as a cross-check and is selected by `par.deriv = 'circle'`. It is not the algorithm: its radius is a tuned quantity, and it is subject to a radius–roundoff trade-off—too small a radius amplifies rounding by $m!/h^m$, too large a one approaches the nearest singularity of Ψ—that the series route does not have. Where both are accurate they agree, which is what makes it useful as a check.

OA.3 Quadrature

All outer integrals are mapped to $(0,1)$ or $(-1,1)$ and evaluated by Gauss–Legendre rules; the periodic ψ-integral in the mean of the t-ratio uses the midpoint rule, which is spectrally accurate for periodic integrands. The maps use the scales $c_B = \mathbb{E}[y'By]$ and $c_C = \mathbb{E}[y'Cy]$, which are available in closed form. With $q, q' \in (0,1)$ and $x \in (-1,1)$ denoting the Gauss–Legendre abscissae, the maps and node counts are:

$$
\begin{aligned}
b: &\quad s = \frac{q}{c_B(1-q)}; && && 64 \text{ nodes},\\
R^2: &\quad s = \frac{q}{c_B(1-q)}, && t = \frac{q'}{c_C(1-q')}; && 48 \times 48,\\
\text{mean of } t: &\quad v = \frac{q}{c(\psi)(1-q)}, && c(\psi) = \cos^2\psi\, c_B + \sin 2\psi\, c_A + \sin^2\psi\, c_C; && 48 \times 64,\\
\text{higher } t: &\quad \ell_1 = \frac{q}{\sqrt{c_B}(1-q)}, && \ell_2 = \frac{x}{\sqrt{c_C}(1-x^2)}, && \ell_3 = \frac{q'}{\sqrt{c_C}(1-q')}; \quad 26 \times 34 \times 26,\\
\text{single odd } t: &\quad \lambda = \frac{q}{c_C(1-q)}, && \theta = \frac{\sigma_\theta(\lambda)\, x}{1-x^2}, && \sigma_\theta(\lambda) = \left(\frac{1}{2\lambda c_B} + \frac{c_C}{c_B}\right)^{1/2}; \quad 64 \times 96.
\end{aligned}
$$

The last map belongs to `tratmom_odd.m`, which computes a single odd moment of $t(\hat{\beta})$ from the two-dimensional representation of the paper; the width of the inner Gaussian ridge in θ grows like $\lambda^{-1/2}$, so the θ scale must be set per λ node—a fixed scale fails badly at small λ. The moments exist for $p < T-1$ (b and $\hat{\beta}$), $p < T-2$ ($t(\hat{\beta})$), and all p (R^2); the programs refuse to compute nonexistent moments.

OA.4 Cost and validation

The table reports wall-clock seconds for a single call, in MATLAB R2025b on a desktop PC with an AMD Ryzen 9 5950X processor, at $\phi = 0.95$, $\rho_{uv} = -0.856$ and $\tilde{\beta} = 0$. Each column delivers every moment up to the order shown.

T	$\mathbb{E}[b]$	$\mathbb{E}[b^4]$	$\mathbb{E}[R^2]$	$\mathbb{E}[(R^2)^2]$	$\mathbb{E}[t]$	$\mathbb{E}[t^4]$
120	0.002	0.003	0.024	0.030	0.004	0.286
960	0.001	0.003	0.026	0.025	0.007	0.274
7 680	0.002	0.003	0.022	0.022	0.003	0.316
61 440	0.002	0.004	0.033	0.031	0.004	0.343
491 520	0.004	0.008	0.035	0.033	0.006	0.355

Every column is flat across a four-thousand-fold increase in the sample size; what residual drift there is comes from memory traffic, not from arithmetic. The two orders of magnitude between the cheapest and the dearest column are the node counts of Section 3 above: 64 evaluations for the moments of b, 48×48 for R^2, 48×64 for the mean of the t-ratio, and $26 \times 34 \times 26$ for its higher moments; when derivatives of order two or more are needed, the work per node grows with the length $p+1$ of the series the jets carry, roughly quadratically

in p. For a single odd moment in isolation, `tratmom_odd.m` needs about a quarter of the evaluations of the full sweep and runs three to four times faster. For contrast, the banded $O(T)$ evaluator of Section OA.1 takes 5.1 seconds for $\mathbb{E}[R^2]$ at $T = 120$ and 179 seconds at $T = 3{,}840$, doubling with each doubling of T. The validation suite (`test_moments.m`) checks (i) both mgf evaluators against a dense determinant at small T (agreement to 3×10^{-15}); (ii) the exact closed forms in the central case $\rho_{uv} = 0$, where $t(\hat{\beta}) \sim t_{T-2}$ and $R^2 \sim \text{Beta}(\frac{1}{2}, \frac{T-2}{2})$ regardless of ϕ (agreement to 2×10^{-7} or better through the fourth moment); (iii) the first-order bias approximation $\mathbb{E}[b] \approx -\rho_{uv}(1+3\phi)/T$ of Stambaugh (1999); and (iv) direct simulation with 2,000,000 replications at $T = 30$, $\phi = 0.95$, $\rho_{uv} = -0.856$, $\tilde{\beta} = 0.1$, where every computed moment lies within 1.4 simulation standard errors of its simulated counterpart. A separate script (`test_oddmom.m`) validates the odd-moment route: the s-derivative jets of `qfmgf_djs.m` against forty-digit dense references (agreement to 4×10^{-14}), and `tratmom_odd.m` against `tratmom.m` at eight configurations spanning $T = 14$ to 960 (agreement between 3×10^{-7} and 4×10^{-6} at the default node counts, improving to 7×10^{-8} on finer grids).

OA.5 The density of R^2 at the origin

This section derives equations (43)–(45) of the paper. Throughout, $\nu = T - 2$ and $w \in (0, 1)$ denotes the argument of the density of R^2.

The exact map from the t-ratio. By (28) of the paper, $t(\hat{\beta}) = \nu^{1/2}\mathcal{R}_1/(\mathcal{R}_2 - \mathcal{R}_1^2)^{1/2}$ and $R^2 = \mathcal{R}_1^2/\mathcal{R}_2$, so

$$\nu + t(\hat{\beta})^2 = \nu\left(1 + \frac{\mathcal{R}_1^2}{\mathcal{R}_2 - \mathcal{R}_1^2}\right) = \frac{\nu\mathcal{R}_2}{\mathcal{R}_2 - \mathcal{R}_1^2}, \qquad \frac{t(\hat{\beta})^2}{\nu + t(\hat{\beta})^2} = \frac{\mathcal{R}_1^2}{\mathcal{R}_2} = R^2, \tag{OA.3}$$

an identity that holds pathwise. The map $t \mapsto t^2/(\nu + t^2)$ carries $\mathbb{R}$ onto $[0, 1)$ and is two-to-one, with the two preimages of w being $\pm w_\nu$, $w_\nu = (\nu w/(1-w))^{1/2}$. Since

$$\frac{\mathrm{d}w_\nu}{\mathrm{d}w} = \frac{1}{2}\left(\frac{\nu w}{1-w}\right)^{-\frac{1}{2}} \frac{\nu}{(1-w)^2} = \frac{\sqrt{\nu}}{2\sqrt{w}\,(1-w)^{\frac{3}{2}}}, \tag{OA.4}$$

the change-of-variables formula for a two-to-one map gives

$$f_{R^2}(w) = \left[f_{t(\hat{\beta})}(w_\nu) + f_{t(\hat{\beta})}(-w_\nu)\right]\frac{\mathrm{d}w_\nu}{\mathrm{d}w} = \frac{\sqrt{\nu}\left[f_{t(\hat{\beta})}(w_\nu) + f_{t(\hat{\beta})}(-w_\nu)\right]}{2\sqrt{w}\,(1-w)^{\frac{3}{2}}}, \tag{OA.5}$$

which is (43). As a check, at $\rho_{uv} = 0$ and $\tilde{\beta} = 0$, where $t(\hat{\beta}) \sim t_\nu$, (OA.5) reproduces the $\text{Beta}(\frac{1}{2}, \frac{\nu}{2})$ density exactly.

The rate at the origin. Let $w \downarrow 0$ in (OA.5). Then $w_\nu \downarrow 0$ and $(1-w)^{3/2} \to 1$, and $f_{t(\hat\beta)}$ is continuous at the origin—it is given by the absolutely convergent integral of Proposition 1 of the paper, whose integrand is continuous in the evaluation point—so the bracket tends to $2f_{t(\hat\beta)}(0)$ and

$$\sqrt{w}\, f_{R^2}(w) \;\longrightarrow\; \sqrt{\nu}\, f_{t(\hat\beta)}(0), \tag{OA.6}$$

which is (44). The limit is finite, and it is strictly positive because $f_{t(\hat\beta)}(0) > 0$: conditional on the predictor, the numerator of the t-ratio has a nondegenerate normal distribution, so the event that Mz and Mr are orthogonal carries positive density. The singularity is of order $w^{-1/2}$ and hence integrable, but it is present for every T, every ρ_{uv} and every $\tilde\beta$.

Closed form when $\rho_{uv} = 0$. Let $\rho_{uv} = 0$ and condition on $Q = z'Mz$. By Section 2.4 of the paper, $t(\hat\beta)$ is then a noncentral t with ν degrees of freedom and noncentrality $\delta = \tilde\beta\sqrt{Q}$. Write it as $(Z+\delta)/(V/\nu)^{1/2}$ with $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_\nu$ independent. Given V, the statistic is normal with mean $\delta(V/\nu)^{-1/2}$ and variance $(V/\nu)^{-1}$, so its density at the origin is $(V/\nu)^{1/2}e^{-\delta^2/2}/\sqrt{2\pi}$; taking the expectation over V,

$$f_{t(\hat\beta)}(0 \mid Q) = e^{-\frac{1}{2}\delta^2}\,\frac{\mathbb{E}[(V/\nu)^{1/2}]}{\sqrt{2\pi}} = e^{-\frac{1}{2}\tilde\beta^2 Q}\, f_{t_\nu}(0), \tag{OA.7}$$

the central density scaled by $e^{-\delta^2/2}$. Since $f_{t_\nu}(0) = 1/(\sqrt{\nu}\,B(\frac{1}{2},\frac{\nu}{2}))$, averaging over Q and substituting into (OA.6) gives

$$\sqrt{w}\, f_{R^2}(w) \;\longrightarrow\; \frac{\mathbb{E}\left[e^{-\frac{1}{2}\tilde\beta^2 Q}\right]}{B\left(\frac{1}{2},\frac{\nu}{2}\right)}. \tag{OA.8}$$

Finally $Q = z'Mz$ with $z \sim \mathcal{N}(0_T, H)$, so its Laplace transform is $\mathbb{E}[e^{-\theta Q}] = |I_T + 2\theta HM|^{-1/2}$; at $\theta = \tilde\beta^2/2$ this is $|I_T + \tilde\beta^2 HM|^{-1/2}$, which is (45). At $\tilde\beta = 0$ the determinant is one and the coefficient reduces to $1/B(\frac{1}{2},\frac{\nu}{2})$, the value implied by the Beta$(\frac{1}{2},\frac{\nu}{2})$ null distribution.

The determinant in closed form. The determinant in (45) is not merely $O(1)$-computable by the machinery of Section 4.4 of the paper; it collapses to an explicit scalar in the primitives of Lemma 2, so that $f_{t(\hat\beta)}(0)$ itself is available in closed form when $\rho_{uv} = 0$. Since $|H| = (1-\phi^2)^{-1}$,

$$\left|I_T + \tilde\beta^2 HM\right| = |H|\left|H^{-1} + \tilde\beta^2 M\right| = \frac{|N|}{1-\phi^2}, \qquad N = H^{-1} + \tilde\beta^2 M. \tag{OA.9}$$

Now H^{-1} is tridiagonal with diagonal $(1, 1+\phi^2, \ldots, 1+\phi^2, 1)$ and off-diagonal $-\phi$, and $M = I_T - 1_T 1_T'/T$, so

$$N = D - \phi^2\left(e_{1,T}e_{1,T}' + e_{T,T}e_{T,T}'\right) - \frac{\tilde\beta^2}{T}1_T 1_T', \tag{OA.10}$$

where D is the $T \times T$ symmetric tridiagonal Toeplitz matrix with

$$p = 1 + \phi^2 + \tilde\beta^2, \qquad q = -\phi. \tag{OA.11}$$

This is exactly the structure of Lemma 2 of the paper, corrected on the two corners and by one rank-one term. Let $|D|$ and $d_1, \ldots, d_4$ be the primitives of that lemma at (p, q) with $m = T$. Applying the matrix determinant lemma to the corner correction, whose two columns are $e_{1,T}$ and $e_{T,T}$, and then to the rank-one term, and using persymmetry ($1_T' D^{-1} e_{1,T} = 1_T' D^{-1} e_{T,T} = d_2$), gives

$$\left|D - \phi^2(e_{1,T}e_{1,T}' + e_{T,T}e_{T,T}')\right| = |D| \left[\left(1 - \phi^2 d_3\right)^2 - \phi^4 d_4^2\right], \tag{OA.12}$$

$$1_T' \left(D - \phi^2(e_{1,T}e_{1,T}' + e_{T,T}e_{T,T}')\right)^{-1} 1_T = d_1 + \frac{2\phi^2 d_2^2}{1 - \phi^2(d_3 + d_4)}, \tag{OA.13}$$

the second because $(1, 1)'$ is an eigenvector of the 2×2 correction matrix. Hence

$$|N| = |D| \left[\left(1 - \phi^2 d_3\right)^2 - \phi^4 d_4^2\right] \left\{1 - \frac{\tilde{\beta}^2}{T}\left[d_1 + \frac{2\phi^2 d_2^2}{1 - \phi^2(d_3 + d_4)}\right]\right\}, \tag{OA.14}$$

and therefore, explicitly,

$$f_{t(\hat{\beta})}(0) = \frac{1}{\sqrt{\nu}\, B\left(\frac{1}{2}, \frac{\nu}{2}\right)} \left(\frac{1 - \phi^2}{|N|}\right)^{\frac{1}{2}} \qquad (\rho_{uv} = 0). \tag{OA.15}$$

Every ingredient of (OA.14) is a closed form in the two roots $r_\pm$ of $x^2 - px + q^2 = 0$, so (OA.15) is an explicit function of T, ϕ and $\tilde{\beta}$ alone, evaluable in $O(1)$ operations. At $\tilde{\beta} = 0$ one has $p = 1 + \phi^2$, the rank-one term drops, and (OA.14) returns $|N| = |H^{-1}| = 1 - \phi^2$, so that (OA.15) reduces to the central t_ν density at the origin, as it must.

For $\rho_{uv} \neq 0$ no such closed form is available, and the obstacle is the conditional mean rather than the conditional variance. Conditioning on the predictor leaves $\mathbb{E}[Mr|z] = \tilde{\beta}\, Mz + \rho_{uv} Mv$, which for $\rho_{uv} \neq 0$ is no longer parallel to Mz, so the residual sum of squares in the denominator of $t(\hat{\beta})$ carries a noncentrality of its own: the statistic is a *doubly* noncentral t rather than a noncentral one, its density at the origin involves a confluent hypergeometric function of a noncentrality that itself depends on z, and nothing closed survives the average over z. The conditional variance is not what blocks it: $I_T - \rho_{uv}^2 G H^{-1} G' = (1 - \rho_{uv}^2) I_T + \rho_{uv}^2 e_{T,T} e_{T,T}'$ departs from a multiple of the identity only in the last coordinate—the one innovation whose partner is not among $z_0, \ldots, z_{T-1}$—and that departure alone would be removable by a single extra scalar integration. We therefore take $f_{t(\hat{\beta})}(0)$ from the triple integral of Proposition 1 of the paper—one evaluation, at $O(1)$ cost per node.

References

Lyness, J. N., Moler, C. B., 1967. Numerical differentiation of analytic functions. SIAM Journal on Numerical Analysis 4, 202–210.